

History of the PCB

Ever thought how the PCB came to be?
Head over here to learn all about it.
<http://bit.ly/pcbhist>

PCB etching

Instead of using breadboards or handmade PCBs, etch your own for a more compact electrical circuit

Mithun Mohandas

readersletters@thinkdigit.com

In the coming months, we'll be concentrating on a few DIY projects that would look really messy and complicate the wiring if we stick to the usual perforated PCB (Printed Circuit Board). Hence, we'll start using EAGLE (Easily Applicable Graphical Layout Editor) and make PCB CAD drawings and then etch them onto copper-plated PCBs. There are multiple methods to do the same and we'll be using the Toner Transfer method. Here is a simple tutorial on how to make your own PCBs.

Things you'll need

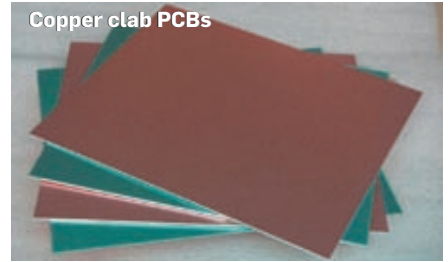
1. Copper-clad PCB of any dimension
2. Laser printer
3. Glossy paper
4. Cloth iron
5. Few grams of Ferric Chloride (FeCl_3) or Hydrochloric Acid (HCl)
6. Hydrogen Peroxide (H_2O_2)
7. Acetone
8. Emery/sand paper (2000 grain) or steel wool or any fine abrasive cloth
9. Plastic container bigger than the PCB
10. Nail varnish
11. Steel/plastic tongs
12. Latex gloves

Steps to follow

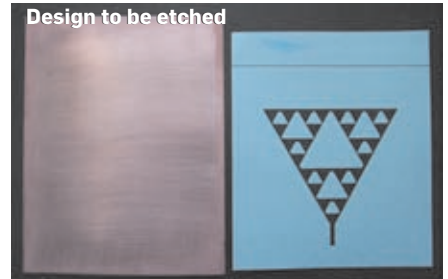
1. Use EAGLE to create your PCB CAD file. The interface is pretty easy to use.
2. Export the CAD file as a PDF or image file.
3. Print the CAD file onto the glossy paper using the Laser printer. This method will only work with laser printers and the basic principle is to transfer the toner onto the PCB.
4. You must keep the copper-clad board free of any fingerprints. If there are any, then use acetone to clean your PCB. If needed, use the emery paper to roughen up the entire surface and wipe with the acetone again.
5. Align the glossy paper with the printout face down onto the copper-plated side of the PCB. Use masking tape to ensure that the paper won't move.
6. Iron the paper now so that the toner transfers from the printout onto the PCB. You'll have to set the cloth iron to a high temperature.
7. Let the board cool down before peeling off the paper. Use a little water to rub off the paper if needed.
8. Ensure that the traces can be easily seen, if any are damaged then wipe clean with acetone and redo the entire process.
9. In the plastic container prepare a mixture of Ferric Chloride solution and Hydrogen Peroxide (1:2 ratio by volume).
10. Immerse the board in the solution and observe as the parts of the board not covered by the toner are eaten away by the solution.
11. Using FeCl_3 takes considerably more time to etch compared to HCl . Depending on the strength of the acid, the etching may complete 3 minutes or 30 minutes.
12. Remove the board from the solution once the exposed copper portion has dissolved and wash it down. Wipe with acetone and check whether all the traces are proper.
13. Solder all the components and then use the nail varnish to cover up the exposed copper to prevent an oxide layer from forming on them. You may tin the exposed copper using solder as well. Enjoy your custom PCB!

Important: The above mentioned chemicals are pretty harmful, avoid inhaling its fumes or coming in direct contact. Carry out the etching part in an open-air environment. Wear gloves at all times, if the molarity of the acid is too high, even gloves would be useless and they would etch too fast. 

Copper clad PCBs



Design to be etched



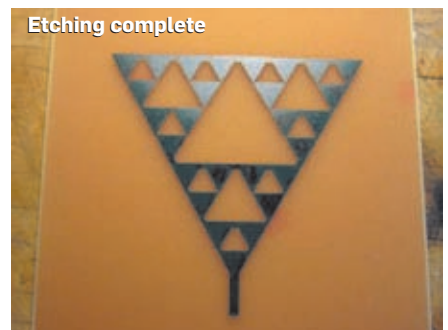
Transfer complete



Etching in progress



Etching complete



Toner removed

